

University of Illinois at Chicago
Department of Chemical Engineering

Syllabus
ChE 201 THERMODYNAMICS FALL 2017

Course Description: Work and energy; conversion of energy; theory of gases and other states of matter; applications to energy conversion devices. Second Law of thermodynamics, entropy, and equilibrium, with applications.

Lecture: 11:00 – 11:50 AM: MWF
Room SES 138
Instructor: Dr. Alan Zdunek, Department of Chemical Engineering, UIC
Office: 223 CEB
Phone: 312-996-4607
Email: zdunek@uic.edu
Office Hours: Monday 1-2pm, Wed 1-2pm

Teaching Assistants:
Ali Davariashtiyani (adavar2@uic.edu) and Jannice Lee (jlee548@uic.edu)

Discussion periods: Tuesday/Thursday: 9:30 – 10:20 AM
Burnham Hall BH B6
Office Hours: To be determined

Text: *Fundamentals of Engineering Thermodynamics*, 8th ed., Moran, Shapiro, Boettner and Bailey 2014 (Hardcover, Looseleaf, Ebook can be used, MUST BE 8th Edition)

Aims, Scope, Level

The engineering science courses that form the basis of chemical engineering can be roughly divided into two sequences: transport processes and thermodynamics. This course is the first in the thermodynamics sequence and is designed to give the student a firm grounding in energy and entropy concepts. Later courses will deal in more detail with phase and reaction equilibrium. Knowledge of thermodynamics is needed in subsequent undergraduate chemical engineering courses. If a student can master the fundamentals early in the course of study, this knowledge will be of benefit throughout the rest of the curriculum as well as in a professional engineering career.

It is assumed that the students in the course have the necessary background in elementary mathematics, and this mathematical background will be used. Also, students should understand mechanics which is usually taught in the first college physics course.

By the end of the term, the students will understand energy and entropy concepts for closed systems and open systems (control volumes). Student will be able to apply engineering methodology to a variety of situations in order to solve thermodynamic problems. Energy and entropy are conceptually difficult and require time and effort on the student's part before they are well understood. Students should read the appropriate sections of the text before lectures, homework assignments should be started as soon as

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possible, and thoughtful questions should be brought up during lecture and discussion.
Good luck!

Student Learning Outcomes covered:

1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Course Outline

- I. Introduction
- II. Fundamentals (Chapter 1)
- III. The First Law (Chapter 2)
- IV. Properties of Pure Fluids (Chapter 3)
- V. Control Volume Analysis (Chapter 4)
- VI. The Second Law of Thermodynamics (Chapter 5)
- VII. Entropy (Chapter 6)
- VIII. Property Relations (Chapter 11 Section 1 through 5)
- IX. Power Cycles
 - A. Rankine and Gas Power Cycle (Chapter 9 Sections 5 through 8)

Exams, Homework and Grading

There will be five or six 'pop' quizzes, two midterm exams, and one final exam. All exams are required. There will be three 50 minute-lectures per week (Mon-Wed-Fri) and one "discussion session" (Tues or Thurs). New material will be covered in lecture. Students should stay current with the material and be prepared to answer questions. During the discussion sessions, homework problems will be worked and occasionally new material covered. The teaching assistant is under no obligation to grade homework that is late, illegible, or sloppy. Scoring fewer than half of the total homework grade is grounds for failing the course. The homework is an important exercise necessary for learning the material and should be worked with care.

Grades:

Two exams (20% each)	40%
Final exam	30%
Quizzes	15%
Homework problems	15%

Additional Information and Policies

1. EXAMS will be held during the class period. Makeup exams will not be provided except for health and weather-related emergencies,
2. Cheating on homework, quizzes, and exams will not be tolerated and appropriate action will be taken including (but not limited to) a failed grade or lowering of the final grade earned in the course.
3. Unannounced 'pop' quizzes will be held periodically during the class period. There will be no makeup quizzes.

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4. Late Homework Policy: Homework can be turned in 1 lecture period late without excuse for up to 50% credit.
5. Homework format guidelines are given in Blackboard/Course Content.
6. Homework/quiz grading: Lowest homework and quiz grade will be dropped.
7. Lecture Mobile Technology Policy will be given in 1st Lecture period.